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Single Qubit Quantum Systems

Multiple Qubit Quantum Systems: Direct/Tensor Product, Entanglements, Gates and Quantum Parallelism

📄 Lab: Interactive Notes on Multi-Qubit Quantum Systems 1h

📺 Video: Multi-Qubit Quantum States 29 min

📝 Graded Assignment: Multiple Qubit Quantum States Submitted

📺 Video: Multi-Qubit Quantum Gates 28 min

📝 Graded Assignment: Multiple Qubit Quantum Gates Submitted

📺 Video: Quantum Parallelism 14 min

📺 Video: No Cloning Theorem 5 min

📝 Graded Assignment: Quantum Parallelism and Implementing Classical Gates 45 min

Bell's Inequality and An Entanglement Game

Grover's Search Algorithm

Problem Set: Week 2

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Quantum Parallelism and Implementing Classical Gates

Next item →

1. Suppose we have a unitary operation U that transforms $|00\rangle$ into $|11\rangle$ and $|11\rangle$ into $|00\rangle$ but leaves the states $|01\rangle$ and $|10\rangle$ unchanged. 3 / 3 points

🔍 Review Learning Objectives

Select all the correct answers from the list below.

Assignment details

👑 Applying U to the state of the form $a|00\rangle + b|11\rangle$ yields the superposition $b|00\rangle + a|11\rangle$

👑 Correct Correct

Submitted May 2, 2:18 AM IST

Try again

👑 The unitary matrix U will be

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Your grade To pass you need at least 80%. We keep your highest score. 100%

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👑 Correct Correct

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❑ The unitary matrix U will be

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

👑 Applying U to the uniform superposition $|\psi\rangle$ of two qubits will leave it unaltered.

👑 Correct Correct

2. We have a two qubit quantum state and an extra ancillary qubit: $|b_1b_2\rangle \otimes |b\rangle$ 3 / 3 points

We would like to implement the AND-gate that transforms

$$|b_1b_2\rangle \otimes |b\rangle \rightarrow |b_1b_2\rangle \otimes |b \oplus (b_1 \wedge b_2)\rangle$$

where $\oplus$ denotes the XOR operation and $\wedge$ denotes the and operation.

Which of the following operations will achieve this?

👑 The multicontrol-X gate with two control qubits is also called the "Toffoli" gate.

👑 Correct

Yes, you can google this one!!

❑ Apply a controlled X gate with b_1 as the control and b as the target, and then apply a controlled X gate with b_2 as the control and b as the target.

👑 Apply a multiccontrolled X gate with b_1, b_2 as the control gates operating on the ancillary qubit b . Note that multi-controlled X gate will apply the X gate onto b only if b_1, b_2 are both 1 or otherwise leave the result unchanged.

👑 Correct

❑ Check if b_1, b_2 are both 1 and just for that component, apply the XOR of the ancillary qubit b with 1.

3. Suppose we have a three qubit system $|b_1b_2b_3\rangle$ and we would like to implement the operation 3 / 3 points

$$|b_1b_2b_3\rangle \longrightarrow |b_1(b_1 \oplus b_2)(b_1 \oplus b_2 \oplus b_3)\rangle$$

where $\oplus$ refers to the XOR operation over bits without using any ancillary qubits. Select the correct option from the list below.

👑 This can be done by first applying a multi-controlled X (Toffoli) gate on b_3 with b_1, b_2 as control qubits and then applying a controlled X gate on b_3 with b_1 as the control qubit.

❑ This operation is not reversible and hence not unitary.

❑ This cannot be achieved without having two ancillary qubits.

❑ Applying the operation twice results in the original state: it is its own inverse.

👑 Correct Correct

4. Suppose we have a quantum circuit implementation of a Boolean function $f(b_1, \dots, b_n)$ that transforms 3 / 3 points

$$|b_1 \dots b_n\rangle \otimes |b\rangle \longrightarrow |b_1 \dots b_n\rangle \otimes |b \oplus f(b_1, \dots, b_n)\rangle$$

What happens if we apply this quantum circuit to the uniform superposition of n qubits and measure the ancillary qubit?

❑ A singularity happens and the world comes to an end.

❑ The ancillary qubit collapses to either a 1 or a 0 and the input qubits $b_1, \dots, b_n$ are unaltered.

❑ The superposition is unaltered if we just measure the ancillary qubit. We have to measure the input qubits to cause the superposition to collapse to a single pure state.

❑ The ancillary qubit collapses to either a 1 or a 0 and the input qubits $b_1, \dots, b_n$ collapse to an arbitrary input $|b_1 \dots b_n\rangle$ for which $f(b_1, \dots, b_n)$ matches the measured result.

👑 The ancillary qubit collapses to either a 1 or a 0 and the input qubits $b_1, \dots, b_n$ collapse to a uniform combination of all those inputs for which $f(b_1, \dots, b_n)$ matches the measured result.

👑 Correct Correct

5. Suppose we implement a quantum and circuit 3 / 3 points

$$|b_1b_2\rangle \otimes |b\rangle \rightarrow |b_1b_2\rangle \otimes |b \oplus (b_1 \wedge b_2)\rangle$$

with ancillary qubit $|b\rangle$ and apply it to the uniform superposition of inputs $|b_1b_2\rangle$ but with $b = |0\rangle$.

Next we measure $|b\rangle$. Which of the following statements are true?

👑 If we measure $b = 1$ the input qubits collapse into $|11\rangle$

👑 Correct

👑 The measurement of $|b\rangle$ yields $|0\rangle$ with $\frac{3}{4}$ probability and $|1\rangle$ with $\frac{1}{4}$ probability.

👑 Correct

❑ Measuring b has no effect on the input qubits.

👑 If we measure $b = 0$ the input qubits collapse into the super position

$$\frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$$

👑 Correct